END TERM EXAMINATION

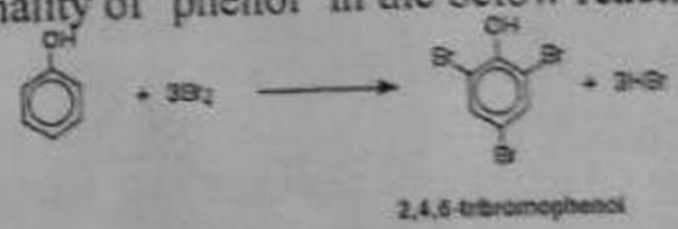
(B. Tech : AI&ML/ CSE-AI)

JULY, 2023)

Subject Code: BAS 106	**Subject:** Environmental Science
Time : 3 Hours	**Maximum Marks :** 60

Note: Q. 1 is compulsory. Attempt one question from each of the units I, II, III & IV. Various scientific symbols have their usual meanings.

Q1	**Attempt all the questions**	**(2.5X 8=20)**
	(a) Chemists sometimes refer to 'by products rather than waste'. List one advantage and one disadvantage of using this term.	
	(b) How the chemical structure and knocking characteristics are related to each other?	
	(c) What is meant by EIA? Why it is necessary to have EIA before the development of any project?	
	(d) What is the role of photosensitisers and photo stabilisers, in degradation of polymers? Give suitable examples	
	(e) Write down the biochemical effects of Chromium?	
	(f) The four (4) R's to be followed for managing solid waste. Explain them.	
	(g) Why is octane number of 2,3 dimethyl pentane higher than that of n-heptane?	
	(h) Explain the causes of soil erosion and enumerate the methods of prevention of soil erosion.	
	UNIT-I	
Q2		**(5,3,2)**
	(a) Distinguish between surface mining and sub surface mining?	
	(b) Discuss in brief the major food problems. How is undernourishment different from malnourishment? (c) Desertification	
Q3		**(5,3,2)**
	(a) According to your point of view, which type of sustainable approaches should be implemented towards water resource management in urban area? (b) Discuss the impacts of Dams on the forests and tribal people.	
	(c) How can you differentiate the onsite impacts caused by the agriculture with that of the off-site impacts?	
	UNIT-I I	
Q4		**(5,5)**
	(a) Discuss on the sources, effects and control of 'Oxides of Sulphur'	
	(b) Write a short note on 'Green Starting Material' in concept of Green chemistry. Write down the green synthesis of adipic acid from D glucose.	

Q5		(4,4,2)
	(a) What is the 'atom economy' of the reaction below for extracting iron? $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$	
	(b) What a brief note on the chemical processes involved in the treatment of hazardous waste technologies?	
	(c) What is the difference between Landfills and sanitary landfills?	
	UNIT-III	
		(2.5, 2.5 ,5)
Q6	(a) What is Biodiesel? What are its advantages over other fuels? (b) How the Geothermal energy is produced? Discuss its Demerits.	
	(c) A sample of coal contains: C=90%; H=4%; and Ash=0.5%; the following data were obtained when the above coal was tested in bomb calorimeter: (i) Weight of Coal Burnt= 0.85 g; Wt of water taken= 650g; Water equivalent of bomb calorimeter= 1800g; Rise in Temperature= 3.42 ^{o}C; Fuse Wire Correction= 8.0 Cal; Acid Correction=40.0 Cal; Calculate the Gross and Net Calorific Value of the coal, assuming the latent heat of condensation of steam as 580 cal/g;	
		(5,3,2)
Q7	(a) A sample of gas contains: C=90%; H=6%; S=2.5%; O=1%; and ash= 0.5%; If 25% excess air is used for combustion, find the weight of air actually supplied per m^3 of this gas.	
	(b) Write a short note on Fractional distillation of petroleum, using a suitable diagram? (c) Write a short note on Ocean Thermal Energy Conversion?	
	UNIT-IV	
		(5,5)
Q8	(a) Enumerate various forms of Mercury (Hg). Which of these forms is highly toxic? Describe its sources and Biochemical Effects.	
	(b) What do you mean by Bio- Plastics? Name any two methods of preparation of plastics from Biopolymers?	
Q9	(a) Write a short note on the following terms by giving suitable examples 2.5 marks (i) Antagonism (ii) Synergism (iii) Effective Dose (ED-50) (b) Plot Dose-Response curve and explain Threshold dose? 2.5 marks	
	(c) (i) What is the functionality of 'phenol' in the below reaction? 2 marks  2,4,6-tribromophenol (ii) If polymer sample has population as? 3 marks 10 molecules of molecular mass each =5,000 20 molecules of molecular mass each =7,500 20 molecules of molecular mass each =10,000 25 molecules of molecular mass each =15,000 Calculate its number average and weight average molecule mass of polymer?	

Session-1